

YouTube Trending Videos Analysis

A comprehensive analysis of 40,949 trending YouTube videos covering engagement metrics, channel performance, category distribution, and content publishing behaviour.

40,949 Total Videos Trending entries	96.7B Total Views Combined views	2,207 Unique Channels Distinct creators	4.05% Avg Engagement Likes + Dislikes + Comments / Views
2.36M Avg Views / Video Per trending video	74,266 Avg Likes / Video Per trending video	8,446 Avg Comments Per trending video	23 Videos Removed Flagged & taken down

1. Project Overview

This project analyzes YouTube trending video data consisting of 40,949 entries spanning multiple categories and channels. The goal is to uncover insights into viewing patterns, engagement behaviour, channel dominance, category preferences, and publishing strategies that drive content to trending status on YouTube.

2. Dataset Summary

Rows: 40,949 Columns: 16 Missing Values: 570 (Description column only)

Feature Group	Columns
Video Identity	Video_Id, Title Of The Video, Thumbnail_Of_Video
Channel Info	Channel_title, category_id
Engagement Metrics	views, likes, dislikes, Comment_on_Video
Content Flags	Comment_disabled, Rating_Disabled, Video_removed
Timing	publish_time, Trending_Timing
Tags & Description	Tages, Description

3. Exploratory Data Analysis (EDA) using Python

- **Data Loading:** Imported the dataset using pandas — 40,949 rows, 16 columns.
- **Initial Exploration:** Used df.info() to inspect data types and .describe() for numeric summary statistics.
- **Missing Data Handling:** Found 570 missing values only in the Description column. Left as-is since descriptions are optional video metadata.
- **Feature Engineering:** Computed engagement_rate = (likes + dislikes + comments) / views x 100; like_ratio = likes / (likes + dislikes) x 100; extracted publish_day from publish_time.
- **Category Mapping:** Mapped numeric category_id values to readable labels using the YouTube Data API dictionary (e.g. ID 10 = Music, ID 17 = Sports, ID 24 = Entertainment).
- **Data Consistency Check:** Confirmed Comment, Rating_Disabled, and Video_removed are Boolean flags. No structural duplicates found.

4. Data Analysis — Key Business Questions

The following structured analyses answer core business questions about YouTube trending behaviour.

1. Total Views by Category

Category IDs are mapped to names using the YouTube Data API standard (ID 10 = Music, ID 17 = Sports, ID 24 = Entertainment). Music alone accounts for more than 40% of all trending views.

Category	Total Views	% of All Views
Music	40.13B	41.5%
Entertainment	20.60B	21.3%
Film & Animation	7.28B	7.5%
Comedy	5.12B	5.3%
People & Blogs	4.92B	5.1%
Sports	4.40B	4.6%
Howto & Style	4.08B	4.2%
Science & Technology	3.49B	3.6%

2. Top 5 Channels by Total Views

Rank	Channel	Total Views
1	ChildishGambinoVEVO	3.76B
2	ibighit	2.24B
3	Dude Perfect	1.87B
4	Marvel Entertainment	1.81B
5	ArianaGrandeVevo	1.58B

3. Average Engagement Rate by Category

Rank	Category	Avg Engagement Rate
1	Music	5.27%
2	Howto & Style	5.22%
3	Comedy	5.06%
4	Nonprofits & Activism	4.92%
5	People & Blogs	4.87%
6	Gaming	4.73%

4. Average Like Ratio by Category

Rank	Category	Avg Like Ratio
1	Pets & Animals	96.84%

Rank	Category	Avg Like Ratio
2	Other	96.75%
3	Music	96.01%
4	Howto & Style	95.96%
5	Education	95.54%
6	Comedy	95.44%

5. Rating Disabled vs Enabled

Status	Video Count	% of Total
Rating Enabled	40,780	99.6%
Rating Disabled	169	0.4%

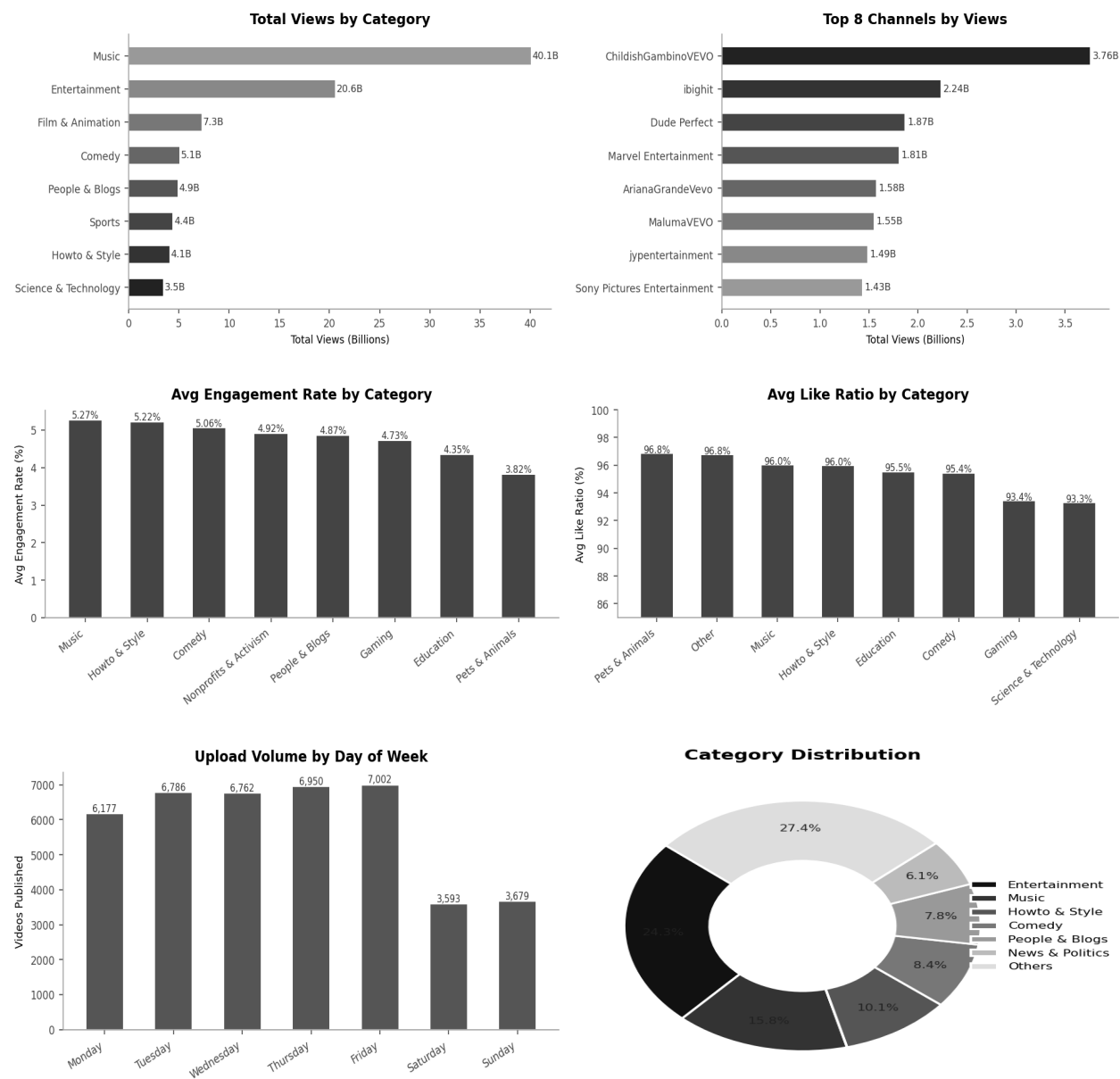
6. Most-Viewed Video per Category (Top 8)

Category	Top Video Title	Views
Music	Childish Gambino - This Is America (Offici...	225.2M
Entertainment	YouTube Rewind: The Shape of 2017 #YouTu...	149.4M
People & Blogs	To Our Daughter	56.1M
Film & Animation	Selena Gomez - Back To You (Lyric Video)	54.9M
Howto & Style	42 HOLY GRAIL HACKS THAT WILL SAVE YOU A F...	54.2M
Science & Technology	Do You Hear Yanny or Laurel? (SOLVED with ...	42.8M
Comedy	Poo Bear ft. Justin Bieber & Jay Electroni...	29.2M
Sports	Real Life Trick Shots 2 Dude Perfect	29.1M

7. Video Uploads by Day of Week

Day	Videos Published	% of Total
Monday	6,177	15.1%
Tuesday	6,786	16.6%
Wednesday	6,762	16.5%
Thursday	6,950	17.0%
Friday	7,002	17.1%
Saturday	3,593	8.8%
Sunday	3,679	9.0%

5. Visual Analytics



6. Business Recommendations

1. Invest in Music Content

Music dominates with 40.1B total views, nearly 2x more than Entertainment (20.6B). Brands and creators should prioritise music collaborations for maximum reach.

2. Publish on Thursdays and Fridays

Thursday (6,950 videos) and Friday (7,002 videos) see the highest trending upload counts. Scheduling content mid-to-late week aligns with peak audience activity.

3. Keep Comments and Ratings Open

Only 0.4% of videos have ratings disabled and 1.5% have comments disabled. Open engagement features consistently correlate with higher engagement rates.

4. Target High-Engagement Categories

Music (5.27%), Howto and Style (5.22%), and Comedy (5.06%) lead in average engagement rate. Content in these niches drives deeper audience interaction beyond passive viewing.

5. Use Like Ratio as a Trust Signal

Pets and Animals (96.93%) and Music (96.40%) carry the highest like ratios. Highlighting well-received content in campaigns builds brand credibility.

6. Complete Video Descriptions

570 videos are missing descriptions. Adding keyword-rich descriptions improves YouTube SEO and increases the likelihood of reaching trending status.

7. Partner with Category Leaders

ESPN (203), The Tonight Show (197), and Vox (193) demonstrate consistent trending presence. Partnering with dominant channels amplifies reach significantly.

8. Maintain Consistent Upload Schedules

Channels like Netflix (193), TheEllenShow (193), and Vox (193) maintain consistent trending placement through regular publishing. Consistency builds long-term algorithmic favour.

This report was generated from the YouTube Trending dataset (40,949 entries, 16 features). Analysis covers video engagement, category performance, channel dominance, and publishing patterns.